Application Layer Case Study: CDNs & Video Streaming

Coverage: Content Distribution Networks (CDNs), DNS Redirection, Edge Server Placement, DASH (Dynamic Adaptive Streaming over HTTP)

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1. CDN Architecture

- Replicates video content across thousands of edge server clusters (PoPs) close to end users.
- DNS Redirection: CNAME DNS intercept directs client to geographically nearest edge cache.

2. DASH Video Streaming

- Video encoded at multiple bitrates and divided into chunks. Client dynamically requests optimal bitrate based on available bandwidth.

■ KTU University High-Yield Exam Questions

- Q: Explain how CDNs use DNS redirection and explain the working of DASH. (8 Marks)